Foundational Model and Generative AI Assignment

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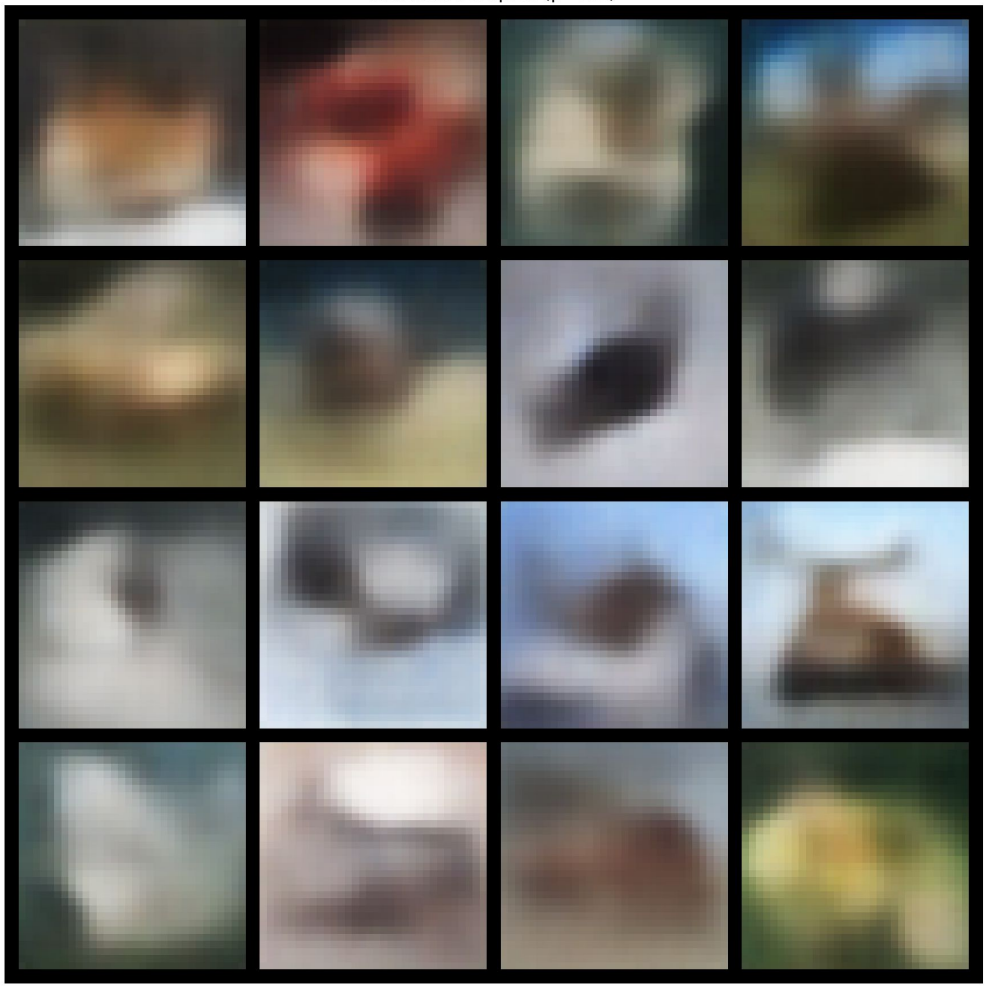
- outputs
  - beta1.0
    - final_random_samples.png
    - interpolation.png
    - model.pt
    - reconstructions_ep1.png
    - reconstructions_ep10.png
    - reconstructions_ep15.png
    - reconstructions_ep20.png
    - reconstructions_ep25.png
    - reconstructions_ep30.png
    - reconstructions_ep35.png
    - reconstructions_ep40.png
    - reconstructions_ep45.png
    - reconstructions_ep5.png
    - reconstructions_ep50.png
    - samples_ep1.png
    - samples_ep10.png
    - samples_ep15.png
    - samples_ep20.png
    - samples_ep25.png
    - samples_ep30.png
    - samples_ep35.png
    - samples_ep40.png
    - samples_ep45.png
    - samples_ep5.png
    - samples_ep50.png
    - training_curves.png

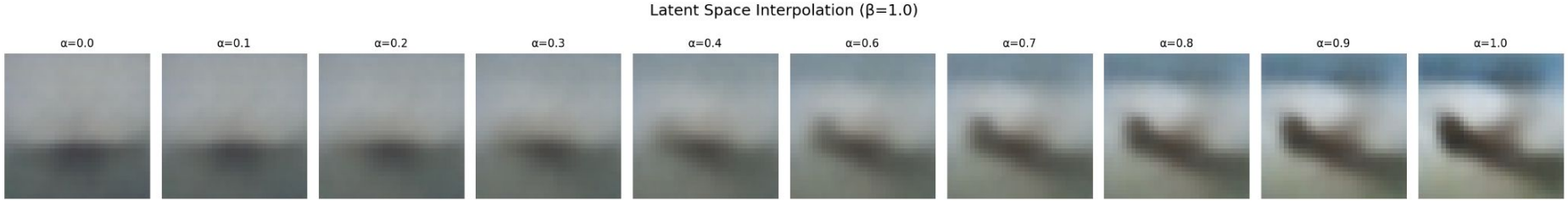
```
Training VAE on CIFAR-10 with latent_dim=64, beta=1.0
Model total parameters: 1,430,979
Trainable parameters: 1,430,979
Epoch 001/50 | Train Loss: 1904.070 | Recon: 1880.985 | KL: 23.085 | Val Loss: 1860.272
Epoch 002/50 | Train Loss: 1849.504 | Recon: 1821.376 | KL: 28.127 | Val Loss: 1845.495
Epoch 003/50 | Train Loss: 1841.903 | Recon: 1811.735 | KL: 30.168 | Val Loss: 1839.251
Epoch 004/50 | Train Loss: 1838.146 | Recon: 1806.775 | KL: 31.371 | Val Loss: 1836.478
Epoch 005/50 | Train Loss: 1835.742 | Recon: 1803.337 | KL: 32.405 | Val Loss: 1833.884
Epoch 006/50 | Train Loss: 1833.995 | Recon: 1801.163 | KL: 32.832 | Val Loss: 1833.745
Epoch 007/50 | Train Loss: 1832.700 | Recon: 1799.415 | KL: 33.285 | Val Loss: 1831.437
Epoch 008/50 | Train Loss: 1831.521 | Recon: 1797.868 | KL: 33.653 | Val Loss: 1832.508
Epoch 009/50 | Train Loss: 1830.661 | Recon: 1796.771 | KL: 33.890 | Val Loss: 1834.311
Epoch 010/50 | Train Loss: 1829.898 | Recon: 1795.735 | KL: 34.163 | Val Loss: 1829.661
Epoch 011/50 | Train Loss: 1829.001 | Recon: 1794.763 | KL: 34.238 | Val Loss: 1828.595
Epoch 012/50 | Train Loss: 1828.788 | Recon: 1794.435 | KL: 34.354 | Val Loss: 1829.674
Epoch 013/50 | Train Loss: 1828.292 | Recon: 1793.792 | KL: 34.501 | Val Loss: 1829.806
Epoch 014/50 | Train Loss: 1827.992 | Recon: 1793.318 | KL: 34.675 | Val Loss: 1830.330
Epoch 015/50 | Train Loss: 1827.620 | Recon: 1792.784 | KL: 34.836 | Val Loss: 1828.270
Epoch 016/50 | Train Loss: 1827.192 | Recon: 1792.332 | KL: 34.860 | Val Loss: 1827.698
Epoch 017/50 | Train Loss: 1826.879 | Recon: 1792.007 | KL: 34.873 | Val Loss: 1829.142
Epoch 018/50 | Train Loss: 1826.579 | Recon: 1791.632 | KL: 34.947 | Val Loss: 1826.988
Epoch 019/50 | Train Loss: 1826.334 | Recon: 1791.242 | KL: 35.092 | Val Loss: 1827.800
Epoch 020/50 | Train Loss: 1825.936 | Recon: 1790.694 | KL: 35.242 | Val Loss: 1826.429
Epoch 021/50 | Train Loss: 1825.457 | Recon: 1790.068 | KL: 35.389 | Val Loss: 1826.290
Epoch 022/50 | Train Loss: 1825.581 | Recon: 1790.013 | KL: 35.567 | Val Loss: 1827.598
Epoch 023/50 | Train Loss: 1825.256 | Recon: 1789.615 | KL: 35.641 | Val Loss: 1826.942
Epoch 024/50 | Train Loss: 1824.688 | Recon: 1788.942 | KL: 35.746 | Val Loss: 1825.601
Epoch 025/50 | Train Loss: 1824.297 | Recon: 1788.553 | KL: 35.744 | Val Loss: 1825.992
Epoch 026/50 | Train Loss: 1824.357 | Recon: 1788.576 | KL: 35.781 | Val Loss: 1826.257
Epoch 027/50 | Train Loss: 1824.424 | Recon: 1788.498 | KL: 35.926 | Val Loss: 1825.458
Epoch 028/50 | Train Loss: 1824.217 | Recon: 1788.233 | KL: 35.984 | Val Loss: 1825.768
```

Final Random Sampling Image

Random Samples (β=1.0)



Interpolation Images



Reconstructed Image Epoch:01

Reconstructed Image on Epochs: 10

Reconstructed Image on Epochs: 15

Reconstructed Image on Epochs: 30

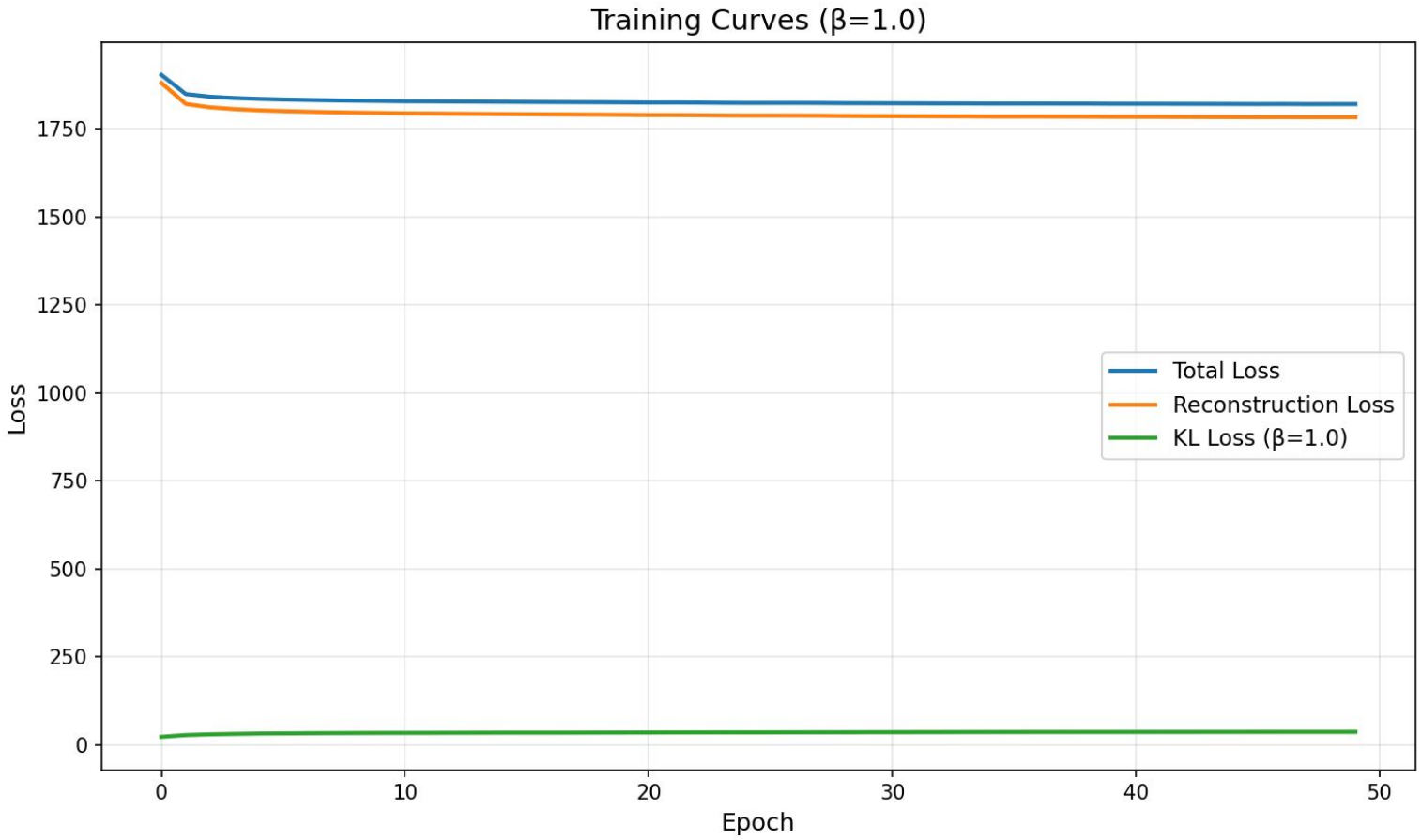
Original vs Reconstructed Images (β=1.0)

Reconstructed Image on Epochs: 40

Original vs Reconstructed Images (β=1.0)

Reconstructed Image on Epochs: 50

Original vs Reconstructed Images (β=1.0)


Training Curves (β=1.0)
Loss
Epoch
Total Loss
Reconstruction Loss
KL Loss (β=1.0)
1750
1500
1250
1000
750
500
250
0
0
10
20
30
40
50

Summary

In this implementation, β directly scales the KL-divergence term in the VAE loss:
Loss = Reconstruction Loss + beta × KL Divergence.
Changing β therefore shifts the balance between reconstruction fidelity and latent space regularization.

When **beta = 1.0**, the model behaves like a standard VAE. The reconstruction term dominates the loss, so the decoder prioritizes producing visually accurate CIFAR-10 reconstructions. The KL term remains relatively small, meaning the latent space is less constrained and allows overlapping encodings. This results in good reconstruction quality but a less disentangled latent representation. Latent space interpolations are smooth but do not strictly separate semantic factors : (colour/shape).

Overall, increasing β improves disentanglement at the cost of reconstruction quality. Lower β favours image fidelity, while higher β encourages a cleaner, more interpretable latent space but risks losing essential visual details.

Comparing Beta =1 vs Beta=5

In this Beta-VAE implementation, Beta controls the strength of the KL-divergence term and therefore governs the balance between reconstruction accuracy and latent space regularization. With **Beta = 1**, the model behaves like a standard VAE, prioritizing reconstruction quality. The decoder receives rich latent information, resulting in sharper and more detailed CIFAR-10 reconstructions. However, the latent space remains loosely structured, with entangled features and overlapping distributions.

Increasing to **Beta = 5** significantly strengthens the KL penalty, forcing the encoder to compress information more aggressively. This produces a more disentangled and well-organized latent space, improving linearity and smoothness in interpolations. However, this comes at the cost of reconstruction fidelity: images become blurrier, with loss of fine details and colour nuances. While Beta = 5 improves interpretability and factor separation, it reduces visual quality and increases the risk of partial posterior collapse.

THANK YOU

for your time and attention.